

Anaerobic Co-digestion of Food Waste and Glycerol in Single-Stage System

Diogenes Theodoros Bussador Matrakas

*Dissertation submitted to Escola Superior Agrária de Bragança to obtain
the Degree of Master in Biotechnological Engineering under the scope of
the double diploma with Universidade Tecnológica Federal do Paraná.*

Supervised by

Dr. Ramiro José Espinheira Martins

Dr. Francisco Menino Destéfanis Vítola

This dissertation does not include the comments and suggestions mentioned by the Jury.

Bragança
September 2026

Acknowledgements

Abstract

The growing generation of food waste and the surplus of crude glycerol from biodiesel production represent two significant organic waste streams requiring sustainable management solutions. This study evaluated the technical feasibility of a single-stage anaerobic co-digestion system treating food waste and glycerol as co-substrates, with the objective of verifying biogas production while providing a valorisation route for crude glycerol. The investigation was conducted in a 1 L continuous stirred-tank reactor operated under mesophilic conditions (35 °C) with a hydraulic retention time of 30 days. Food waste was collected from a university canteen, and crude glycerol was obtained from biodiesel production residues; anaerobic sludge from a municipal wastewater treatment plant served as inoculum. The reactor was operated in sequential phases: an initial adaptation period, co-digestion of food waste and glycerol, glycerol only feeding, and a post-feeding period. Biogas production was monitored continuously using an automatic methane potential test system, while process stability was assessed through measurements of pH, total alkalinity, volatile fatty acids, and their ratio. Substrate characterisation included total solids, volatile solids, density, total carbon, and total nitrogen analyses. Results demonstrated that co-digestion of food waste and glycerol yielded the highest methane production rates and cumulative volumes, with flow spikes immediately following feeding events indicating rapid substrate conversion. In contrast, glycerol-only feeding resulted in reduced biogas production despite continued methanogenic activity. Total alkalinity remained elevated following bicarbonate supplementation, maintaining pH around 7 throughout most of the experiment; volatile fatty acid concentrations fluctuated between 3000–5000 mg CH₃COOH/L, with VFA to alkalinity ratios of 0.3–1.2. These findings confirm the hypothesis that glycerol can be useful for biogas production when co-digested with food waste or alone. The study demonstrates the technical viability of anaerobic co-digestion as a strategy for valorising crude glycerol within a circular bioeconomy framework, though further research is needed to optimise glycerol loading rates and develop robust feeding strategies for industrial application.

Resumo

A crescente geração de resíduos alimentares e o excedente de glicerol bruto proveniente da produção de biodiesel representam duas produções significativas de resíduos orgânicos que requerem soluções sustentáveis. Este estudo avaliou a viabilidade técnica de um sistema de codigestão anaeróbia de uma única etapa, tratando resíduos alimentares e glicerol como co-substratos, com o objetivo de verificar a produção de biogás e, simultaneamente, proporcionar uma via de valorização para o glicerol bruto. A investigação foi conduzida num reator contínuo de tanque agitado de 1 L, operado em condições mesofílicas (35 °C), com um tempo de retenção hidráulica de 30 dias. Os resíduos alimentares foram recolhidos numa cantina universitária e o glicerol bruto foi obtido a partir de resíduos da produção de biodiesel; lamas anaeróbias de uma estação de tratamento de águas residuais municipais serviram como inóculo. O reator foi operado em fases sequenciais: um período inicial de adaptação, codigestão de resíduos alimentares e glicerol, alimentação exclusiva com glicerol e um período pós-alimentação. A produção de biogás foi monitorizada continuamente através de um *automatic methane potential test system*, enquanto a estabilidade do processo foi avaliada por medições de pH, alcalinidade total, ácidos graxos voláteis e respectiva razão. A caracterização dos substratos incluiu análises de sólidos totais, sólidos voláteis, densidade, carbono total e nitrogênio total. Os resultados demonstraram que a codigestão de resíduos alimentares e glicerol originou as taxas de produção de metano e os volumes acumulados mais elevados, com picos imediatamente após as alimentações, indicando uma rápida conversão do substrato. Em contraste, a alimentação exclusiva com glicerol resultou numa produção reduzida de biogás, apesar da atividade metanogénica continuada. A alcalinidade total manteve-se elevada após a suplementação com bicarbonato, mantendo o pH em torno de 7 durante a maior parte da experiência; as concentrações de ácidos graxos voláteis flutuaram entre 3000 e 5000 mg CH₃COOH/L, com razões AGV/alcalinidade de 0,3 a 1,2. Estes resultados confirmam a hipótese de que o glicerol pode ser útil para a produção de biogás quando codigerido com resíduos alimentares ou isoladamente. O estudo demonstra a viabilidade técnica da codigestão anaeróbia como estratégia de valorização do glicerol bruto no âmbito de uma bioeconomia circular, embora seja necessária investigação adicional para otimizar as cargas de glicerol e desenvolver estratégias de alimentação

robustas para aplicação industrial.

Contents

1	Introduction	1
2	Objectives	3
2.1	General Objectives	3
2.2	Specific Objectives	3
3	Literature Review	4
4	Materials and Methods	8
4.1	Collection and Preparation of the Food Waste	8
4.2	Crude Glycerol Collection	8
4.3	Inoculum Collection	8
4.4	Physical and Chemical Characterisation	9
4.4.1	TS and VS Determination	9
4.4.2	Density Determination	11
4.4.3	TOC and TN Determination	11
4.5	Experiment Configuration	11
4.5.1	Feeding Solution	12
4.5.2	Biogas Quantification	13
4.5.3	Analysis Methods	13
5	Results	15
5.1	Reactor Condition and Visual Observations	15
5.2	Biogas Production	15
5.2.1	Overall Production	17
5.2.2	Adaptation Phase	19
5.2.3	Co-digestion of Food Waste and Glycerol	20
5.2.4	Glycerol only Feeding Period	22
5.2.5	Post-feeding Period	24
5.2.6	Alkalinity and Volatile Fatty Acids	26

Contents	vi
5.2.7 Substrate used as feed	27
6 Conclusion	29
References	30

Chapter 1

Introduction

The generation of organic waste has become one of the defining environmental challenges of our time, particularly as urbanization and industrial activity continue to expand. A major contributor is food waste, a global stream that in the European Union alone is estimated at 88 million tonnes per year, excluding inedible parts removed from the supply chain (Scherhaufer et al., 2018). When considering only post-processing activities – wholesale, retail, food service and households – approximately 66.3 million tonnes of food waste are produced annually in the EU-27+1 (Albizzati et al., 2021). The scale of this loss highlights the urgent need for improved management strategies that move beyond simple disposal.

The environmental burden of food waste is substantial. It accounts for roughly 15–16% of the total climate impact of the entire food supply chain, translating to about 186 Mt CO₂-eq of greenhouse gas emissions, alongside significant contributions to acidification and eutrophication (Scherhaufer et al., 2018). The majority of these impacts arise during primary production, making prevention the single most effective strategy; however, when waste cannot be avoided, its valorisation through anaerobic digestion or composting is favoured over landfill disposal (Albizzati et al., 2021; Scherhaufer et al., 2018). Despite this, much food waste continues to be landfilled or incinerated, forfeiting the opportunity to recover its embedded energy and nutrients.

Parallel to the food waste challenge, the rapid expansion of biodiesel production has generated an excess of another organic by-product: crude glycerol. Formed in the transesterification of triglycerides, approximately 10 kg of crude glycerol are produced for every 100 kg of biodiesel (Garlapati et al., 2016). Global biodiesel output, driven by renewable energy mandates, has caused the glycerol market to swell – with crude glycerol supply projected to reach about 6.33 million tonnes by 2025, of which nearly 680 000 tonnes are expected to be highly impure due to the shift towards waste-based feedstocks (Attarbachhi et al., 2023).

Crude glycerol composition varies widely depending on the oil source and catalyst used,

but it typically contains methanol, soaps, free fatty acids, salts and other matter organic non-glycerol (MONG) compounds (Anand & Saxena, 2012; Attarbachhi et al., 2023). These impurities are known to inhibit microbial growth and fermentation, as demonstrated by the severe reduction in 1,3-propanediol production when untreated crude glycerol from jatropha, soybean or sunflower oils was used (Anand & Saxena, 2012). Purification to technical or pharmaceutical grade is energy-intensive and often uneconomical for small and medium biodiesel plants (Attarbachhi et al., 2023), prompting the search for direct, cost-effective valorisation routes. Because of its high organic content and ready biodegradability, crude glycerol has attracted attention as an ideal co-substrate for anaerobic digestion, where it can serve as a concentrated carbon source while being simultaneously detoxified by dilution and microbial synergy (Gebreegziabher et al., 2025; Tomczak et al., 2025).

Anaerobic digestion (AD) is a mature biological process in which consortia of microorganisms convert organic matter into biogas – primarily methane (55–65%) and carbon dioxide – under oxygen-free conditions, passing through the sequential stages of hydrolysis, acidogenesis, acetogenesis and methanogenesis (Gebreegziabher et al., 2025). AD offers a unique combination of benefits: it produces renewable, storable energy that can replace fossil fuels; generates a nutrient-rich digestate that can be used as organic fertiliser; and reduces the environmental load of organic wastes by avoiding landfilling and its associated methane emissions (Gebreegziabher et al., 2025; Scherhauser et al., 2018).

Within this framework, anaerobic co-digestion – the simultaneous treatment of two or more substrates – has emerged as a particularly effective strategy. Co-digesting crude glycerol with nitrogen-rich materials such as animal manure, sewage sludge or food waste balances the carbon-to-nitrogen ratio, dilutes potential inhibitors, provides essential macro- and micronutrients, and fosters synergistic microbial interactions (Gebreegziabher et al., 2025; Tomczak et al., 2025). The result can be a dramatic improvement in biogas and methane yield: studies report increases ranging from 14% to 226% when crude glycerol is added to a variety of base substrates, with methane contents generally exceeding 60% as long as glycerol does not exceed about 8% (v/v) of the feed (Gebreegziabher et al., 2025; Tomczak et al., 2025).

However, excessive glycerol loads can lead to volatile fatty acid accumulation, pH drop and methanogen inhibition, underscoring the need for careful optimisation (Gebreegziabher et al., 2025; Tomczak et al., 2025). Overall, the convergence of abundant food waste and surplus crude glycerol supplies, together with the proven capability of anaerobic digestion to convert these streams into renewable methane and fertiliser, represents a tangible route towards a circular, low-carbon bioeconomy.

Chapter 2

Objectives

2.1 General Objectives

The primary objective of this dissertation project is to evaluate the efficiency of a single-stage anaerobic co-digestion system using organic food waste and glycerol as substrates. The investigation will be conducted by determining biogas production, with a focus on industrial feasibility.

The central hypothesis is that the controlled addition of glycerol, a by-product of the biodiesel industry, enhances biogas production due to its high energy content and rapid biodegradability, provided that adequate operational conditions are maintained to avoid volatile fatty acid overload.

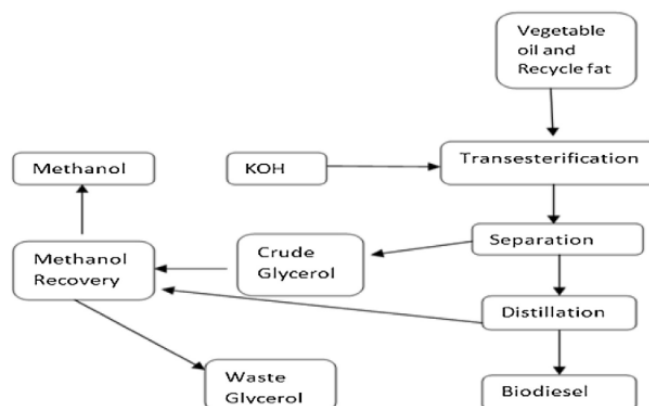
2.2 Specific Objectives

1. Characterize the substrates (food waste and glycerol), evaluating physicochemical parameters relevant to anaerobic digestion.
2. Operate the single-stage anaerobic co-digestion system, assessing process stability.
3. Evaluate the techno-economic feasibility of using glycerol as a co-substrate, considering energy yield and potential for industrial application.

Chapter 3

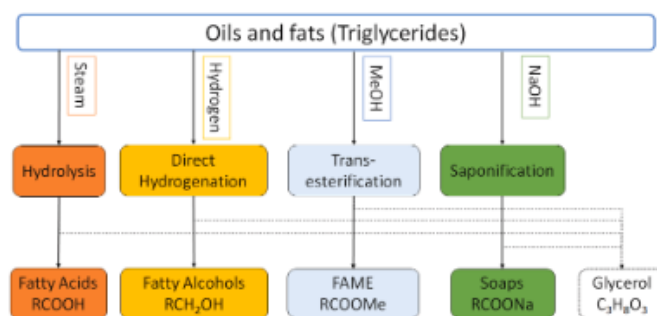
Literature Review

The rapid expansion of biodiesel production has generated an oversupply of crude glycerol, a by-product of the transesterification output, this graph 3.1 shows the steps to produce biodiesel and when glycerol is produced (Garlapati et al., 2016). This waste stream, loaded with impurities such as methanol, soaps, and hydroxides, the graph 3.2 shows the processes and what they contaminate the glycerol with, poses environmental disposal challenges, also with the variation of impurities depending on the source, there's no standard method to use this source of carbon (Anand & Saxena, 2012; Gebreegziabher et al., 2025). Anaerobic digestion (AD) aligns with circular economy principles by converting organic residues into biogas while recycling nutrients, making crude glycerol a promising co-substrate (Gebreegziabher et al., 2025).



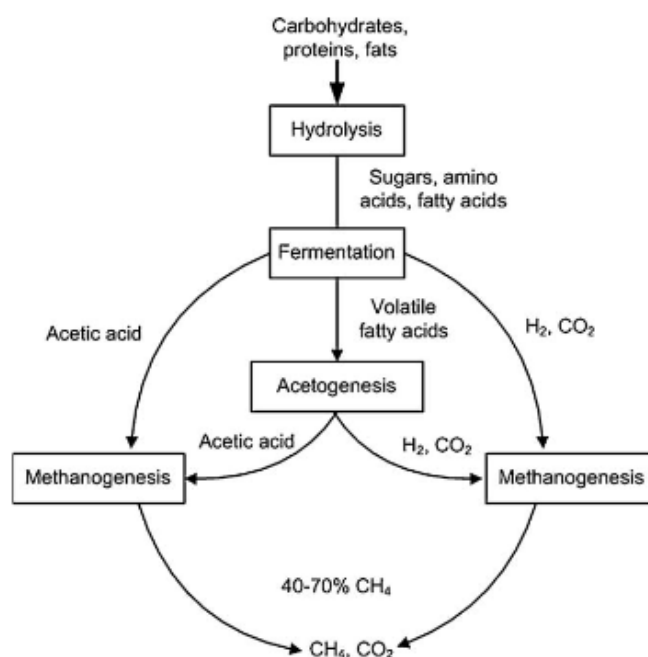
Graph 3.1: Source: Garlapati et al. (2016)

Gebreegziabher et al. (2025) reported that co-digestion of crude glycerol can boost methane production compared to mono-digestion of the base substrate. Tomczak et al. (2025) reviewed one-stage AD experiments and concluded that a methane content above 60% could be consistently obtained when the crude glycerol fraction did not exceed 8% (v/v) of the feed, the flow chart by Li et al. (2011) on the graph 3.3 it shows that using



Graph 3.2: Source: Attarbach et al. (2023)

AD there could be up to 70% methane production. However, the same review emphasized that increasing the glycerol loading beyond a system-specific threshold may provoke inhibitory effects, likely due to the accumulation of volatile fatty acids (VFAs) and the impurities present in crude glycerol (Anand & Saxena, 2012; Tomczak et al., 2025). The heterogeneous composition of crude glycerol, affected by oil feedstock, catalyst, and biodiesel processing, demands careful feedstock characterization and, if necessary, simple pretreatment to avoid process upsets (Attarbach et al., 2023).

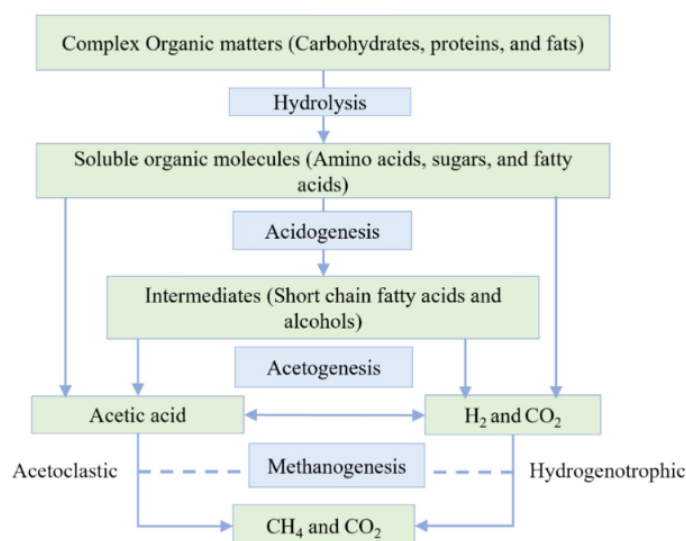


Graph 3.3: Source: Li et al. (2011)

Liquid anaerobic digestion (AD), typically conducted in a continuous stirred tank reactor (CSTR), operates at total solid concentrations of 0.5–15% and is generally used for treating sewage sludge, manure, and food waste (Li et al., 2011). The CSTR's continuous mixing provides uniform distribution of substrates, microorganisms, and heat, which is

essential when feeding a fermentable substrate. Temperature is a critical operational parameter; the activity of methanogenic consortia is strongly temperature dependent, and both mesophilic and thermophilic regimes have been applied to AD (Choorit & Wisarnwan, 2007). Tomczak et al. (2025) noted that the majority of glycerol co-digestion studies have been performed under mesophilic conditions, while thermophilic operation, which could offer higher conversion rates, still requires more systematic investigation. The CSTR platform is well suited to explore such temperature effects because its design allows easy control of heating and hydraulic retention time.

The selection of a robust microbial inoculum is decisive for rapid start-up and stable performance of a CSTR fed with glycerol. Anaerobic sludge from municipal wastewater treatment plants (WWTPs) is the most common source of inoculum because it harbours a diverse consortium of hydrolytic, acidogenic, and methanogenic microorganisms, the flow chart from Gebreegziabher et al. (2025) on the graph 3.4 shows each step of the AD until the methane production. Posmanik et al. (2020) demonstrated that granular sludge, obtained from up-flow anaerobic sludge blanket reactors treating wastewater, outperforms suspended biomass as an inoculum. Although their work focused on batch tests, the findings strongly suggest that using granular sludge from a WWTP as seed for a CSTR could accelerate acclimation, improve process stability, and reduce the risk of acidification during load increases.



Graph 3.4: Source: Gebreegziabher et al. (2025)

When glycerol starts being degraded it makes the AD process susceptible to VFA accumulation and pH drop if the organic loading rate is not properly controlled. Therefore, close monitoring of process indicators is essential. Ribas et al. (2007) evaluated several methods for determining these parameters specifically in anaerobic reactors, highlighting

the importance of accurate VFA and alkalinity measurements for timely detection of process imbalance. Using the method from DiLallo & Albertson (Ribas et al., 2007) provide a validated method for measuring VFAs and alkalinity. Maintaining a favourable VFA-to-alkalinity ratio is particularly critical when co-digesting glycerol, as excess VFAs can inhibit methanogenesis (Tomeczak et al., 2025).

The available literature confirms that crude glycerol is a high-energy substrate that can significantly boost methane production in liquid AD systems. A CSTR provides the mixing and temperature control required to harness this potential. Using granular sludge from a wastewater treatment plant as the inoculum can enhance start-up speed and methanogenic activity, as demonstrated by comparative studies. To unlock the full benefits while avoiding inhibition, the glycerol loading must be kept below documented thresholds, and the reactor state should be continuously tracked through standard measurements of VFAs and alkalinity. Further research is needed to optimise thermophilic CSTR operation and to establish long-term performance data for granular-sludge-inoculated reactors treating crude glycerol.

Chapter 4

Materials and Methods

4.1 Collection and Preparation of the Food Waste

Food waste was collected from the canteen at the Instituto Politécnico de Bragança (IPB), prioritizing residues with high proportions of rice, pasta, and bread to maximize the sugar content and minimize the nitrogen content. The waste was diluted with approximately 2.5 L of distilled water and homogenised using a blender to obtain a liquid paste. It was then stored in a refrigerator at 4 °C to prevent spoilage and suppress microbial and fungal activity (Degueurce et al., 2020).

4.2 Crude Glycerol Collection

A 6 L bottle of crude glycerol, obtained from previous biodiesel production experiments at IPB, was provided for this study. Crude glycerol from biodiesel typically contains impurities such as catalyst residues (Attarbachí et al., 2023), which may influence its behaviour in anaerobic digestion.

4.3 Inoculum Collection

The anaerobic inoculum consisted of sludge collected from Águas do Norte wastewater treatment plant in Bragança. This ensured the availability of a microbial population for the anaerobic digestion (Posmanik et al., 2020).

4.4 Physical and Chemical Characterisation

The total solids (TS), volatile solids (VS), fixed solids (FS), and density were determined for the food waste, crude glycerol, and inoculum. Total organic carbon (TOC) and total nitrogen (TN) were determined for the food waste and crude glycerol.

4.4.1 TS and VS Determination

TS and VS were measured using a sequential gravimetric method based on Standard Methods 2540 B and 2540 E (Baird et al., 2017). The method involves controlled drying and calcination steps to quantify total solids and volatile solids fractions.

Initially, porcelain crucibles were placed in a muffle furnace at 550 °C for 1 h. For cooling, they were transferred to an oven at 105 °C for 15 min and then moved to a desiccator until they reached room temperature; their empty masses were recorded. For each material, a sample was placed into three crucibles and the wet mass was measured. The crucibles were then dried in an oven at 105 °C for 2 h (crude glycerol required 6–8 h to reach constant mass). After drying, the crucibles were cooled in a desiccator to room temperature and weighed again to determine TS.

The same crucibles were subsequently placed in a muffle furnace at 550 °C for 50 min. Afterwards they were transferred to an oven at 105 °C for 15 min and then moved to a desiccator until they reached room temperature; the final masses were recorded to calculate VS.

TS, VS and FS were calculated as follows:

$$TS = \frac{M_{105} - M_{crucible} * 1000}{V_{sample}} \quad (4.1)$$

$$VS = \frac{M_{105} - M_{550} * 1000}{V_{sample}} \quad (4.2)$$

$$FS = \frac{M_{550} - M_{crucible} * 1000}{V_{sample}} \quad (4.3)$$

where M_{wet} is the mass of crucible plus wet sample, M_{105} is the mass after drying at 105 °C, M_{550} is the mass after ignition at 550 °C, $M_{crucible}$ is the mass of the empty crucible and V_{sample} is the volume of the sample. The volume is in *mL* and the weight is in *mg*. The results are on the table 4.1.

Table 4.1: TS, VS and FS results

Sample		M_{crucible}	M_{wet}	M_{sample}	V_{sample}	M_{105}	TS (g/L)	Average TS (g/L)	M_{550}	VS (g/L)	Average VS (g/L)	FS (g/L)	Average FS (g/L)
Food Waste	1	78.7310	105.9875	27.2565	25.5	91.0021	481.0194		79.2024	462.5408		18.4786	
	2	84.3206	113.2979	28.9773	0.0271	98.8684	536.3998	536.3998	84.7435	520.8068	520.8068	15.5930	18.4786
	3	77.5887	108.8399	31.2512	0.0292	93.4417	541.9933		78.1848	521.6134		20.3799	
Glycerol	1	84.2193	88.6434	4.4241	0.0050	88.4966	863.4402		84.2144	864.4293		-0.9891	
	2	73.4780	77.9874	4.5094	0.0050	77.8594	867.7240	867.7240	73.4733	868.6548	868.6548	-0.9308	-0.9891
	3	86.2012	90.6107	4.4095	0.0049	90.5034	871.3421		86.1958	872.4358		-1.0937	
Inoculum	1	78.7310	105.9875	27.2565	0.0255	91.0021	481.0194		79.2024	462.5408		18.4786	
	2	84.3206	113.2979	28.9773	0.0271	98.8684	536.3998	536.3998	84.7435	520.8068	520.8068	15.5930	18.4786
	3	77.5887	108.8399	31.2512	0.0292	93.4417	541.9933		78.1848	521.6134		20.3799	

Source: Author

Table 4.2: Density results

Sample		M_{cylinder}	M_{total}	V_{sample}	Density (g/L)	Average Density (g/L)
Food Waste	1	136.0000	242.0000	0.0992	1068.7905	1068.4376
	2	116.0000	223.0000	0.1002	1068.0847	
Glycerol	1	114.2100	203.4700	0.0999	893.2353	893.0740
	2	115.7400	204.4400	0.0993	892.9127	
Inoculum	1	235.0000	242.0000	0.0992	1068.7905	1068.4376
	2	216.0000	223.0000	0.1002	1068.0847	

Source: Author

4.4.2 Density Determination

Density was determined by a gravimetric–volumetric method using graduated cylinders. Dry cylinders were weighed empty, then filled with distilled water at 20 °C and weighed again to verify the reading accuracy (density of water at 20 °C = 998.21 g L⁻¹). After drying, each cylinder was filled with a known volume of sample and weighed. The density of each material was calculated using

$$\rho = \frac{M_{\text{total}} - M_{\text{cylinder}}}{V_{\text{sample}}} \quad (4.4)$$

where M_{total} is the mass of cylinder plus sample, M_{cylinder} is the empty cylinder mass, and V_{sample} is the measured volume, the masses were measured in g and the volume in L. The results are on the table 4.2.

4.4.3 TOC and TN Determination

Total organic carbon (TOC), total carbon (TC) and total nitrogen (TN) were measured using a Shimadzu TOC-5000A analyser. Prior to analysis, the food waste was diluted 400-fold and the crude glycerol was diluted 800-fold to bring the concentrations within the linear range of the instrument. The results are on the table 4.3.

4.5 Experiment Configuration

The reactor had a useful volume of 1 L and was operated as a continuous stirred-tank reactor (CSTR) under anaerobic conditions. Mixing was provided by a magnetic stir bar, and the overflow exit tube was kept closed with a valve to maintain an anaerobic environment.

Table 4.3: TS, VS and FS results

Sample		TOC (mg/L)	TC (mg/L)	TN (mg/L)	Dilution	TC (mg/L)	Average TC (mg/L)	TN (mg/L)	Average TN (mg/L)
Food Waste	1	5.03900	5.04400	0.68910	400	2017.6	1778.6	275.64	281.76
	2	3.92700	3.84900	0.71970		1539.6		287.88	
Glycerol	1	8.80800	8.81200	0.58750	800	7049.6	5523.2	470	488.6
	2	4.99200	4.99600	0.63400		3996.8		507.2	

Source: Author

The reactor was kept at 35 °C, close to the optimal temperature for mesophilic microorganisms (Choorit & Wisarnwan, 2007). The organic loading rate (OLR) was set with a max of 4 kg VS m⁻³ d⁻¹, with a hydraulic retention time (HRT) of 30 days, conditions for stable co-digestion of food waste and glycerol (Tomczak et al., 2025). The pH was maintained around 7 by incorporating KHCO₃ into the feed solution at a concentration of 10 g L⁻¹, and adding KHCO₃ to the reactor whenever the pH dropped lower than 6.

The organic loading rate (OLR) and hydraulic retention time (HRT):

$$\text{HRT} = \frac{V}{Q} \quad [\text{d}] \quad (4.5)$$

$$\text{OLR} = \frac{S_{\text{in}} \times Q}{V} \quad [\text{kg VS m}^{-3} \text{ d}^{-1}] \quad (4.6)$$

where S_{in} is the influent VS concentration (kg m⁻³), Q is the volumetric flow rate (m³ d⁻¹), and V is the reactor working volume (m³).

4.5.1 Feeding Solution

The feed was designed to achieve a carbon-to-nitrogen (C/N) ratio of 25:1, which is optimal for anaerobic digestion (Li et al., 2011). Characterisation of the food waste and crude glycerol showed that both substrates contained nitrogen, making it difficult to balance the C/N ratio using both of them together. Therefore, pure glycerol was used as an additive to allow precise control of the C/N ratio. Based on the chosen HRT and the daily volumetric flow rate, the reactor was fed once per day to ensure a flow rate high enough to transport the solid food-waste suspension through the feeding tube.

The feed was made with a restriction of needing to have less than 15% of TS%. The amount of OLR and if KHCO₃ was added, was decided by the pH and how was the methane production of the reactor. If it was too acidic, it was used a lower OLR with KHCO₃, if it was at the right amount it was keep the same amount, and if it was too basic it was added a higher OLR without KHCO₃. The feed storage always had some remainig feed, and the

new one was added on top, diluting or concentrating the feed going to the reactor.

The feed was also supplemented with nutrients to complement the feed and create a better feed for the microorganisms. Table 4.4 shows the nutrients used and there ratios.

Table 4.4: Nutrients used and there ratios.

Nutrients	Mass (mg/350 mL)	Mass (mg/L)	Mass (g/L)
MgSO ₄ ·7H ₂ O	18.9000	54.0000	0.0540
FeSO ₄ ·7H ₂ O	522.8000	1493.7143	1.4937
CoCl ₂ ·6H ₂ O	0.4200	1.2000	0.0012
ZnCl ₂	10.5000	30.0000	0.0300
NiCl ₂ ·6H ₂ O	0.0105	0.0300	0.00003
EDTA	0.2100	0.6000	0.0006
CuSO ₄ ·5H ₂ O	9.2300	26.3714	0.0264

Adapted from: Alqudah (2025)

4.5.2 Biogas Quantification

It was used the *bioprocess control AMPTS II (Automatic Methane Potential Test System)* system (fig. 4.1), to measure the gas flow that was comming out of the reactor, with a CO₂ fixing unit using a solution of 3 M NaOH.

4.5.3 Analysis Methods

It was used the method of DiLallo & Albertson (Ribas et al., 2007), to measure the alkalinity and volatile fatty acids. The sample collected was first centrifuged, at 2604 G for 30 minutes, to filter the particulated before doing the measurements.

$$TA = \frac{N_{acid} * V_{acid} * 50000}{V_{sample}} \quad (4.7)$$

$$VFA = \frac{N_{base} * V_{base} * 60000}{V_{sample}} \quad (4.8)$$

The volumes were measured in mL, the sample volume was always 50 mL, the normality (N) of the acid and base, at first was 0.1, but then it was changed to 1. The eq. (4.7) is expressed in mgCaCO₃/L, and the eq. (4.8) is expressed in mgCH₃COOH/L.



Figure 4.1: Source: Author

Chapter 5

Results

5.1 Reactor Condition and Visual Observations

The reactor was inoculated with anaerobic sludge and initially contained a murky, light brown suspension (fig. 5.1a). Within three days of operation, the colour changed to black (fig. 5.1b), which is characteristic of active anaerobic digestion and is typically caused by the precipitation of metal sulphides, particularly iron sulphide, under reducing conditions. This visual change indicated the establishment of a suitable anaerobic environment and the onset of methanogenic activity.

At the end of the experimental period, after approximately two weeks of unattended operation, four discrete masses resembling fungal colonies were observed on the internal surfaces of the reactor during cleaning (figs. 5.2a and 5.2b). The origin of these growths is uncertain, but they may have been introduced via the food waste feed or during periods when the feeding tube was opened for repairs (e.g. leaks) and when changing the feed that the tube had to be cleaned. The presence of such organisms could have competed with the anaerobic microbial community for substrate, potentially affecting biogas production in the later stages of the experiment.

5.2 Biogas Production

Biogas production was monitored continuously using AMPTS II, which recorded both the cumulative volume of methane produced and the instantaneous flow rate. The entire experimental period is summarised in figs. 5.3 and 5.4. To facilitate interpretation, the data have been divided into four distinct phases based on the feeding regime: (i) an adaptation phase 5.2.2, (ii) co-digestion of food waste and glycerol 5.2.3, (iii) glycerol only feeding 5.2.4, and (iv) a post-feeding period 5.2.5.



(a) Murky suspension



(b) Black suspension

Figure 5.1: Visual changes in the reactor.



(a) Masses inside the dry reactor



(b) Masses floating in the reactor

Figure 5.2: Visual for the masses inside the reactor.

5.2.1 Overall Production

The cumulative methane production over the whole experiment (fig. 5.3) shows a continuous increase, indicating that the reactor maintained methanogenic activity throughout the monitored period. The total flow profile (fig. 5.4) exhibits distinct peaks corresponding to daily feeding events, with variations in amplitude reflecting the different substrate compositions and organic loading rates applied during each phase. And a outlier peak at day 53, that was when the feed was changed from a mixture of food waste and crude glycerol, to only crude glycerol, and due to some complications, there was that anomaly.

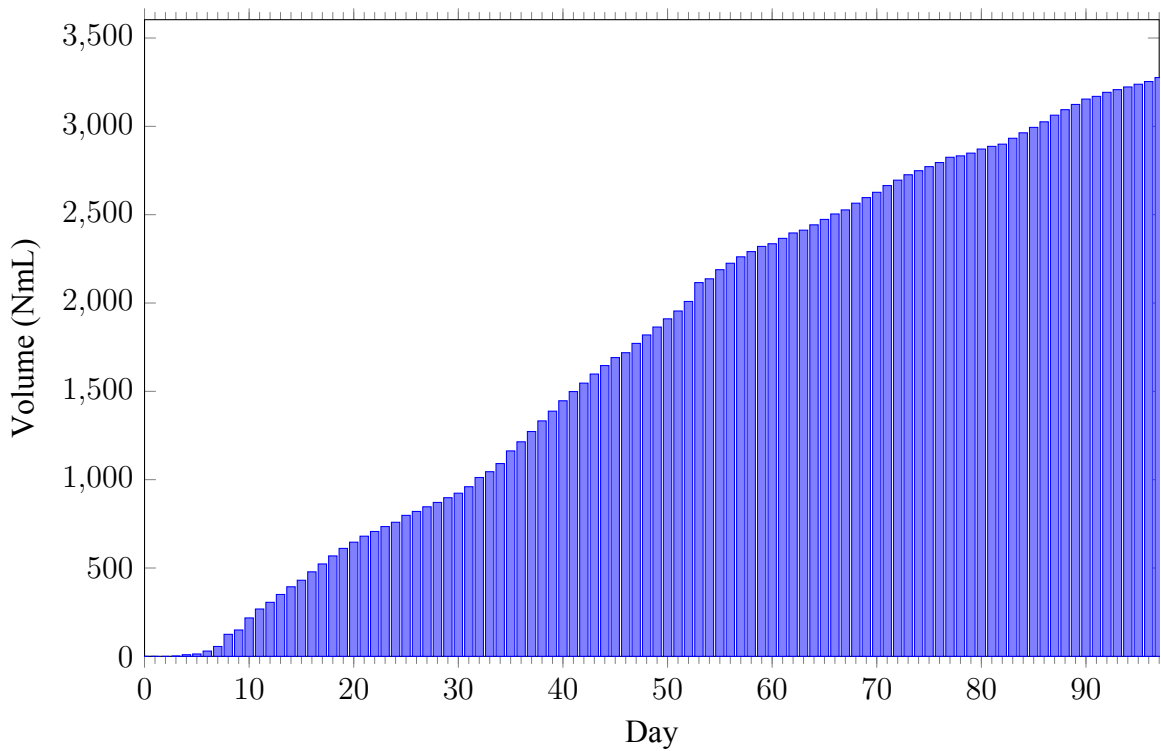


Figure 5.3: Source: Author

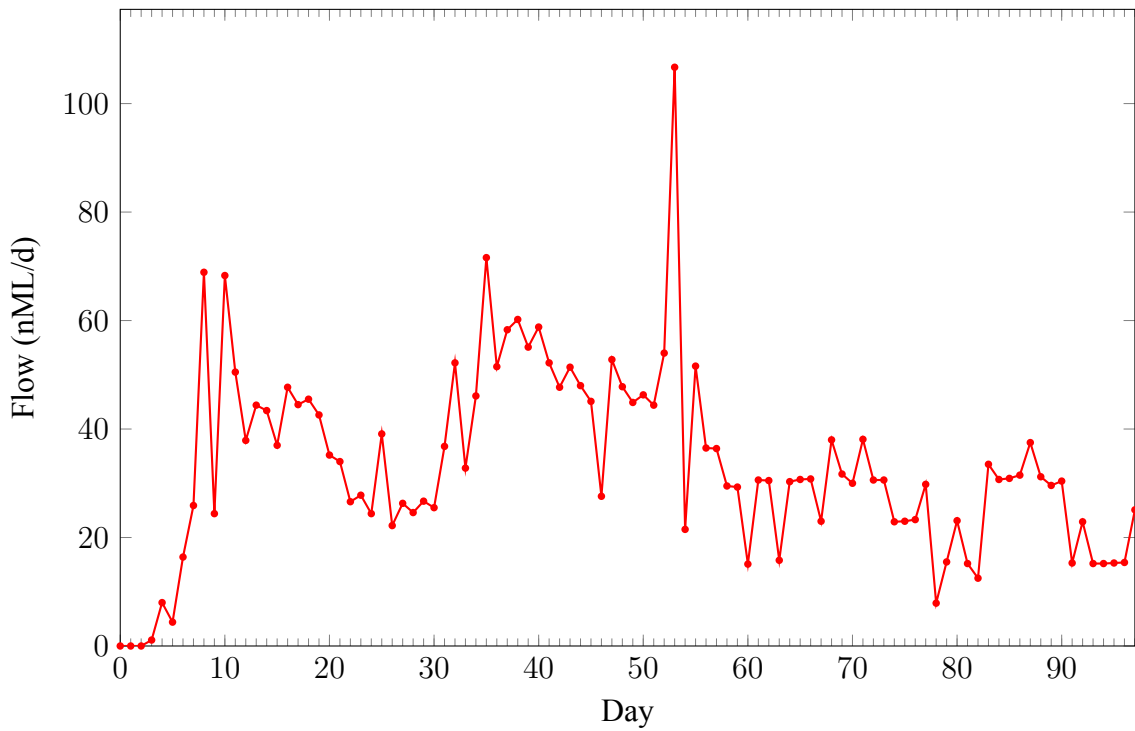


Figure 5.4: Source: Author

5.2.2 Adaptation Phase

During the initial adaptation phase, biogas production was low (figs. 5.5 and 5.6). This lag period is normal for anaerobic reactors as the microbial community adjusts to the new substrate and operating conditions. The small but measurable methane flow confirmed that the viability of the inoculum, and the gradual increase in flow suggested the onset of exponential growth of the microorganisms.

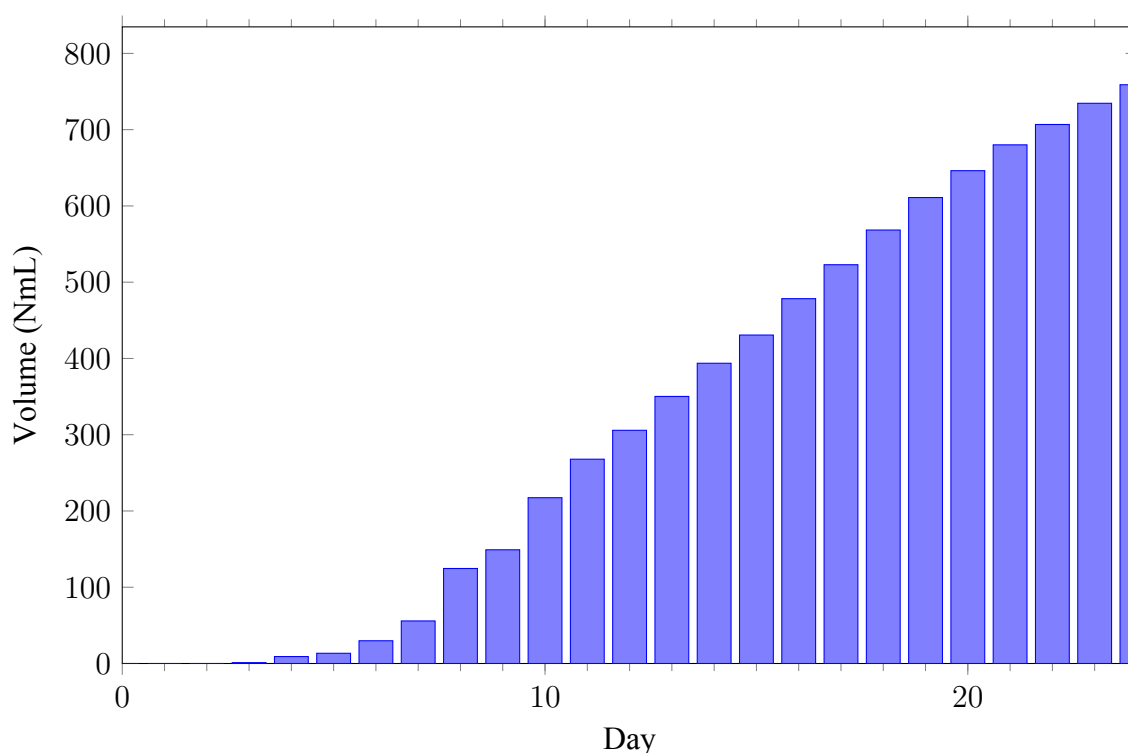


Figure 5.5: Source: Author

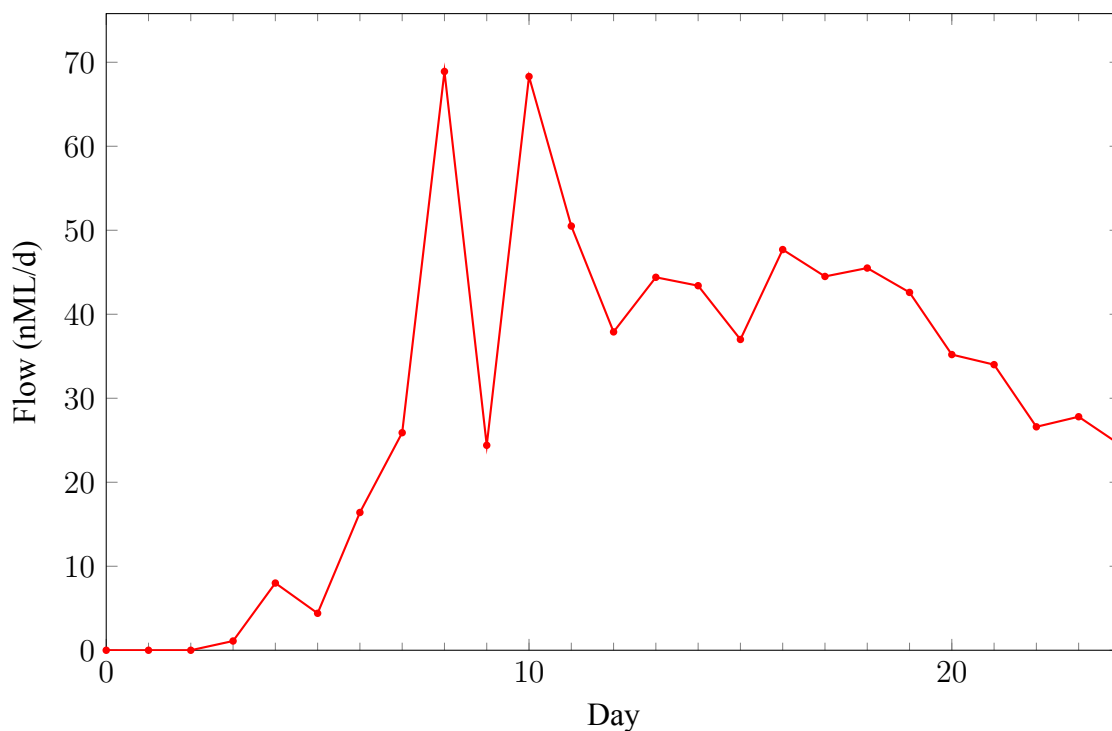


Figure 5.6: Source: Author

5.2.3 Co-digestion of Food Waste and Glycerol

The reactor was fed with a mixture of food waste and crude glycerol with a higher concentration, a marked increase in biogas production was observed (figs. 5.7 and 5.8). The cumulative methane volume rose steeply, and the flow rate showed pronounced spikes immediately after each feeding event (fig. 5.9). These spikes are typical of readily degradable substrates, which are quickly hydrolysed and converted to volatile fatty acids and subsequently to methane. The average flow rate during this phase was higher than in any other phase, indicating that the co-digestion of food waste and glycerol provided a balanced nutrient supply and avoided the inhibition often associated with high glycerol concentrations alone.

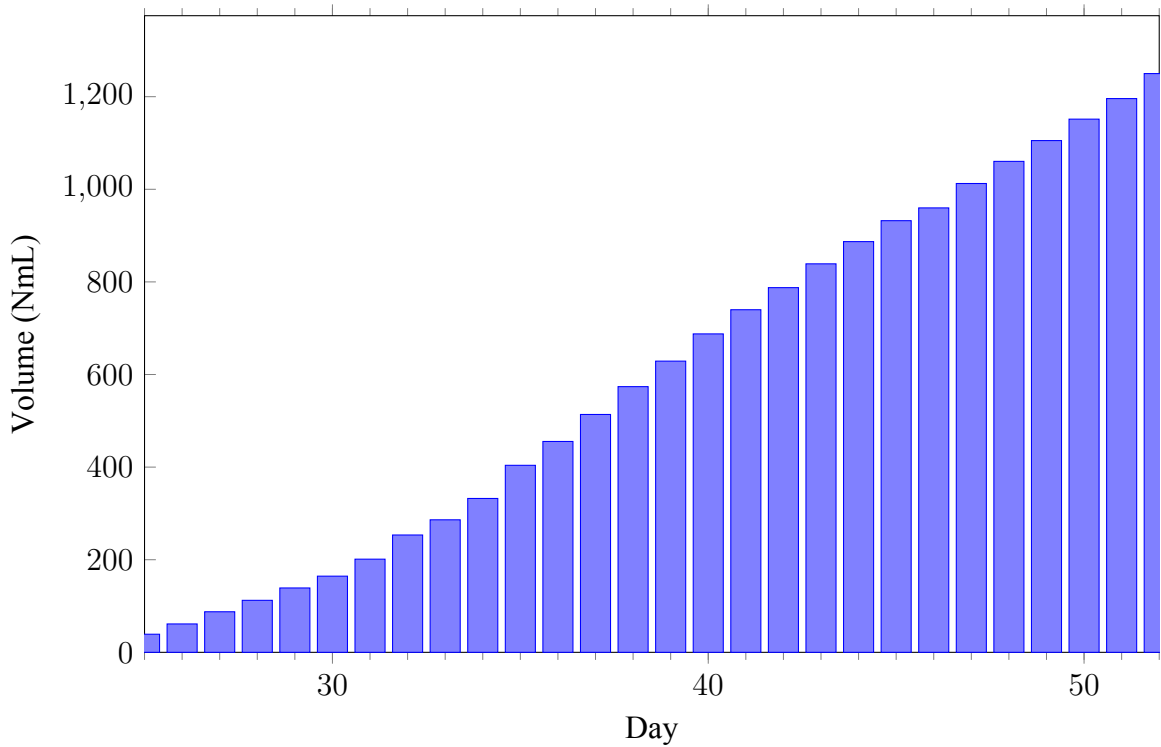


Figure 5.7: Source: Author

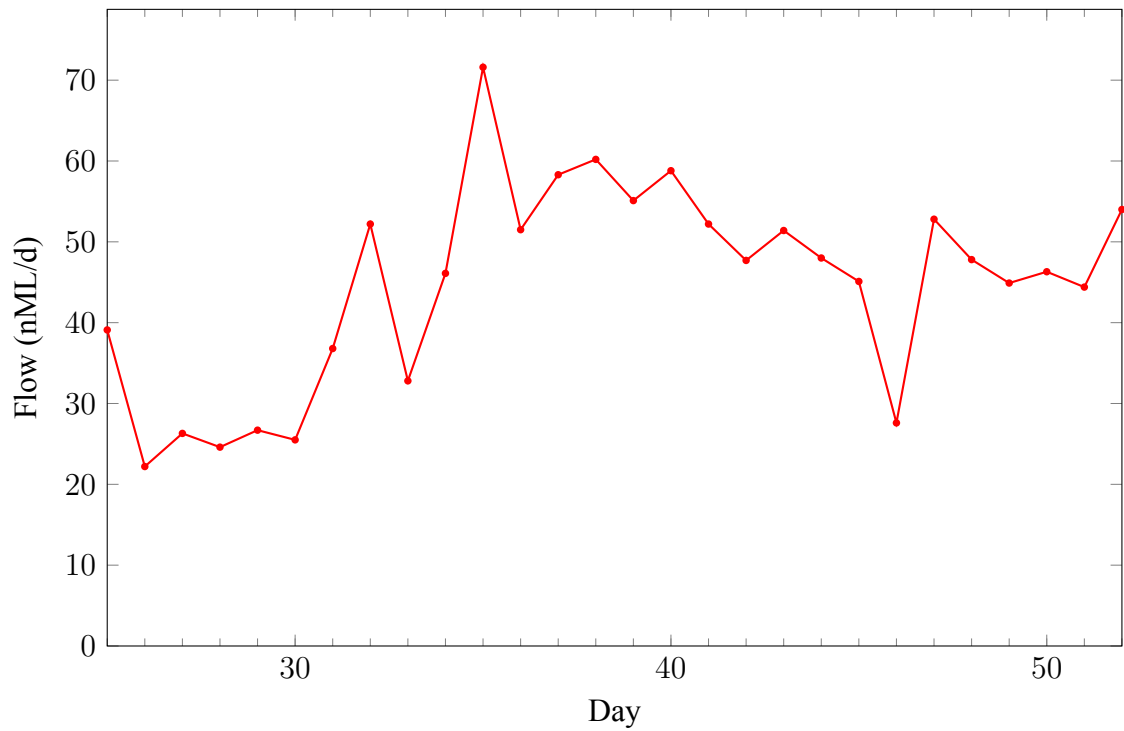


Figure 5.8: Source: Author

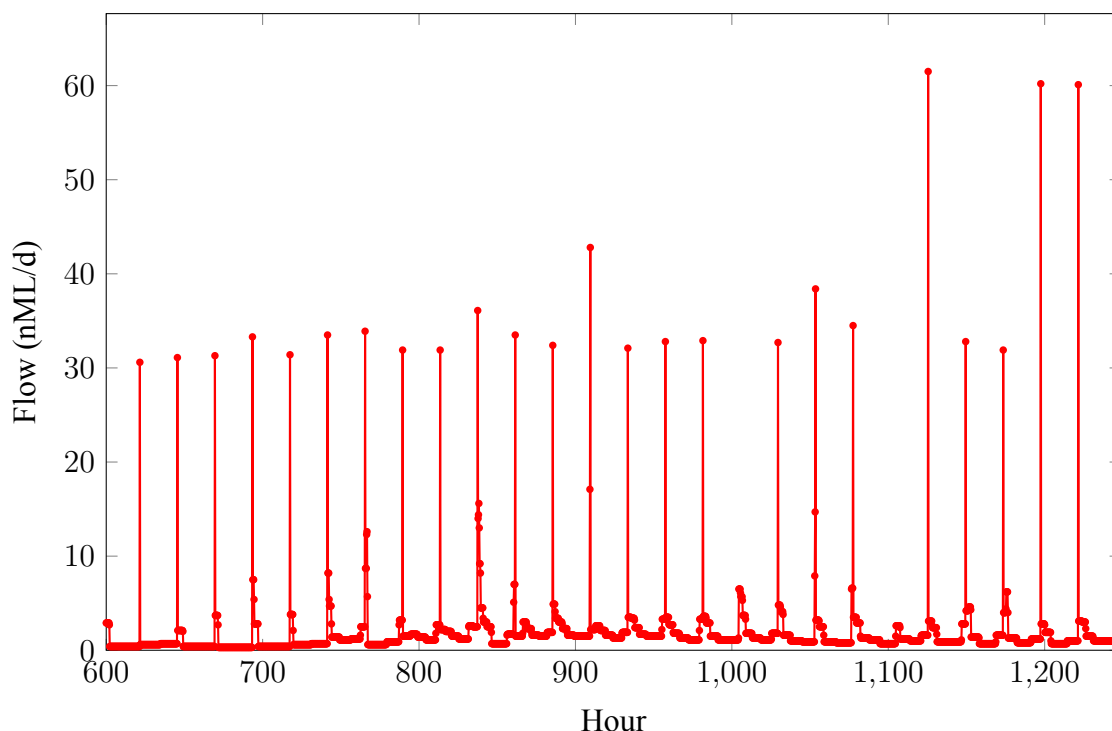


Figure 5.9: Source: Author

5.2.4 Glycerol only Feeding Period

When the feed was changed to crude glycerol alone, biogas production decreased noticeably (figs. 5.10 and 5.11). Although spikes in flow were still observed after feeding (fig. 5.12), their magnitude was bigger than during the co-digestion phase, and the cumulative methane volume plateaued more quickly. Several factors may explain this lower performance:

- The glycerol could have been transformed too fast or too slow, to maintaining a concentration of intermediate products.
- The retention rate of 1/30 days may have been insufficient to wash out inhibitory by-products or to maintain a stable population of glycerol-fermenting bacteria.
- The appearance of fungal masses towards the end of this phase could have further stressed the anaerobic community by competing for substrate or introducing toxic metabolites.

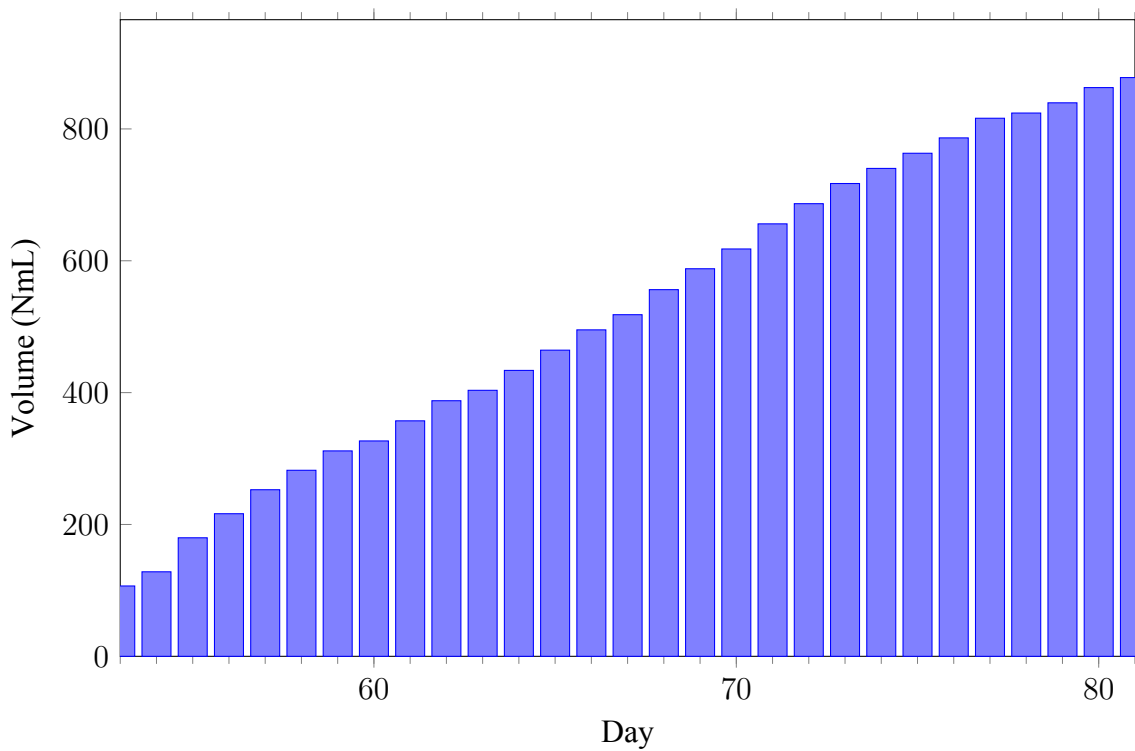


Figure 5.10: Source: Author

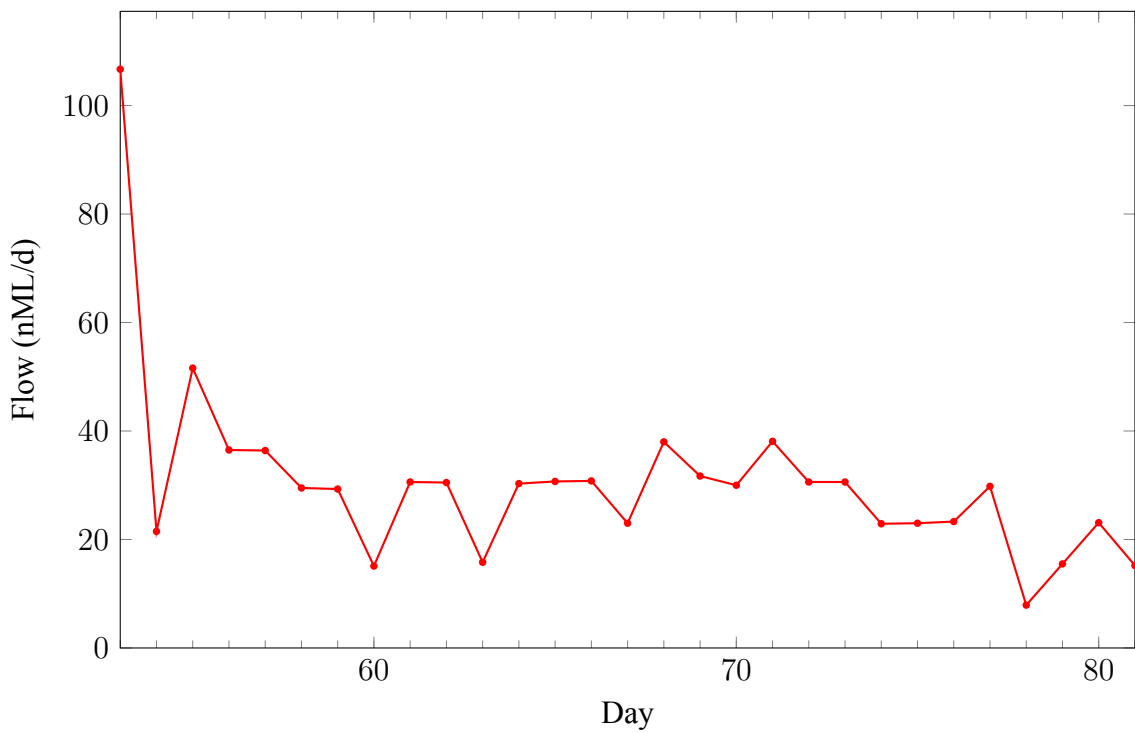


Figure 5.11: Source: Author

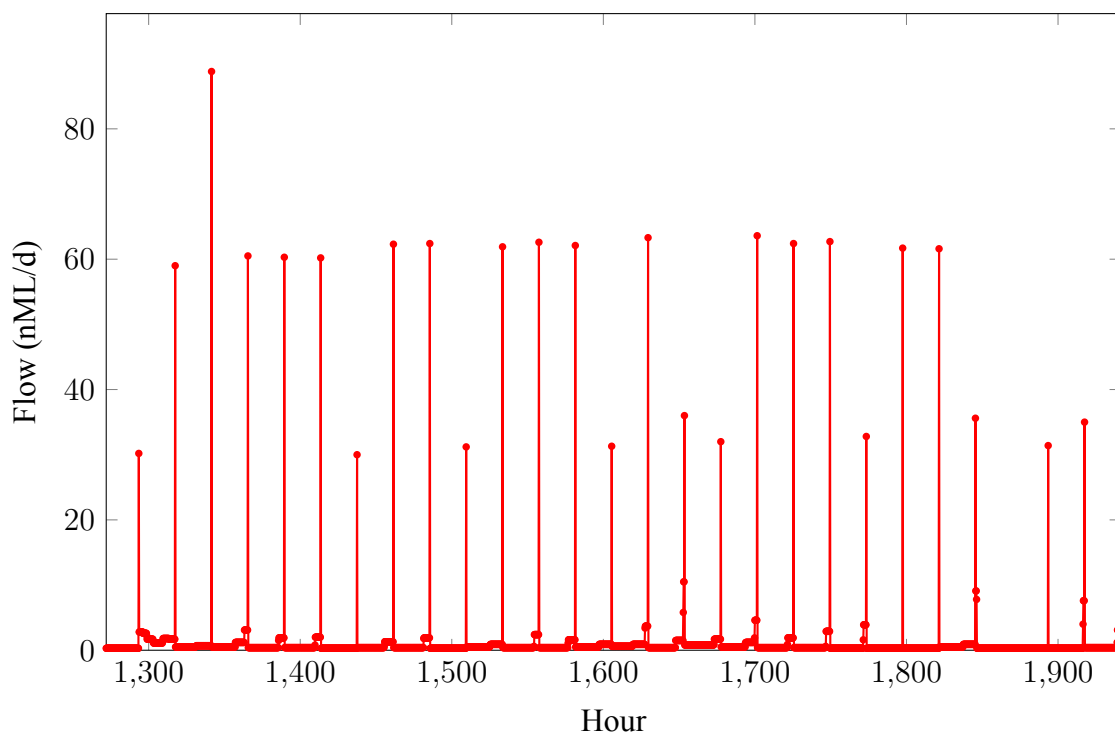


Figure 5.12: Source: Author

5.2.5 Post-feeding Period

During the final two weeks, no fresh substrate was added to the reactor, it just had some residual substrate from the glycerol only phase for a week. Nevertheless, a small residual biogas production was recorded (figs. 5.13 and 5.14). This production can be attributed to the slow degradation of accumulated organic matter and microbial decay products. The flow rate declined gradually, confirming that the reactor was no longer being actively fed and that the remaining substrate was limited.

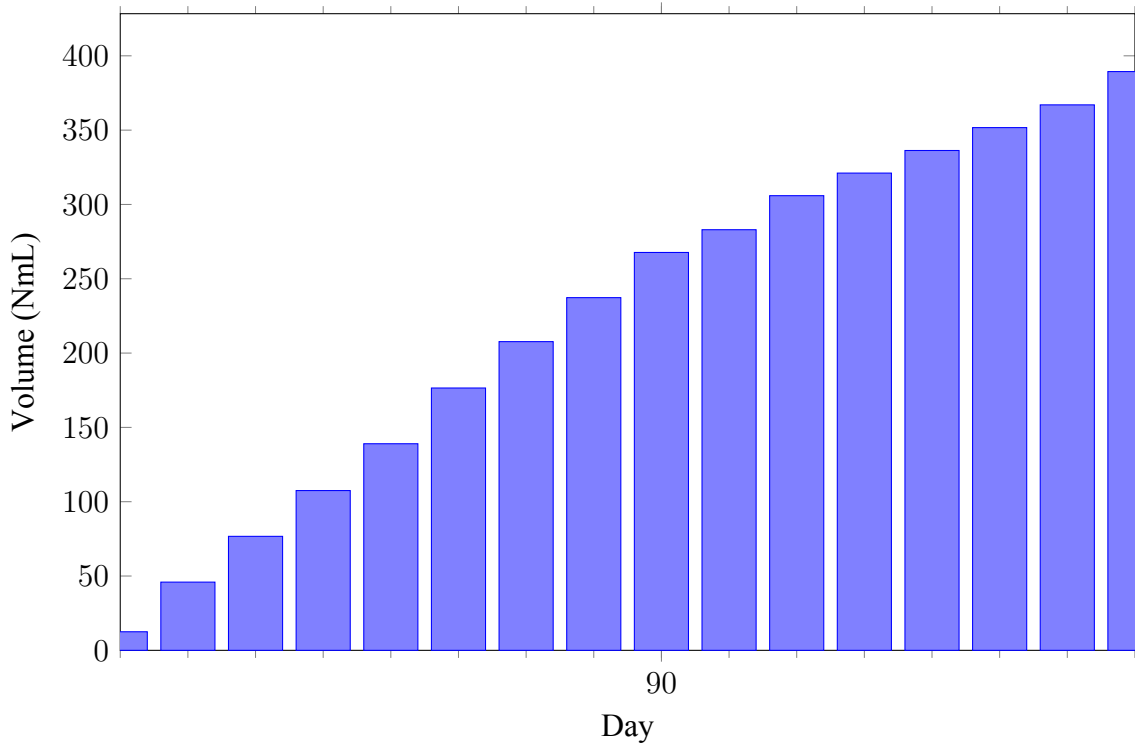


Figure 5.13: Source: Author

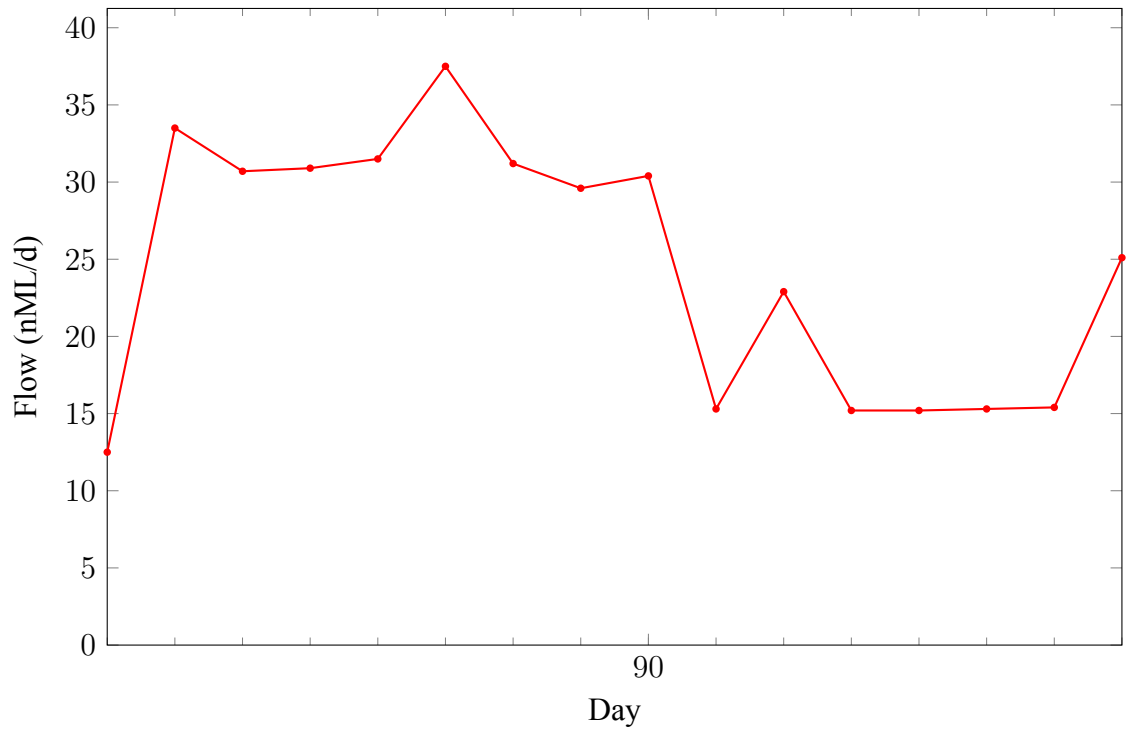
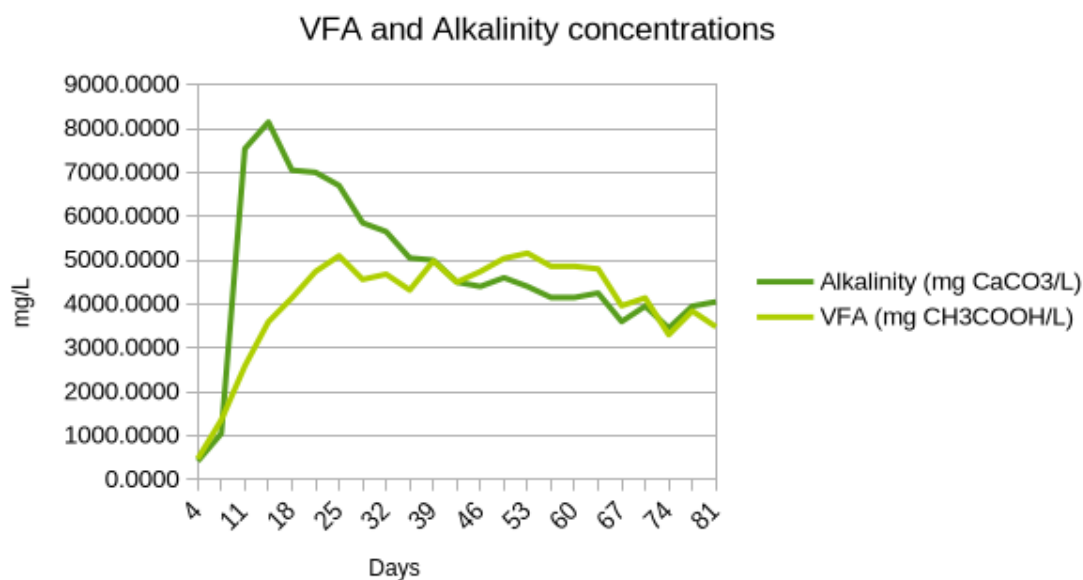


Figure 5.14: Source: Author

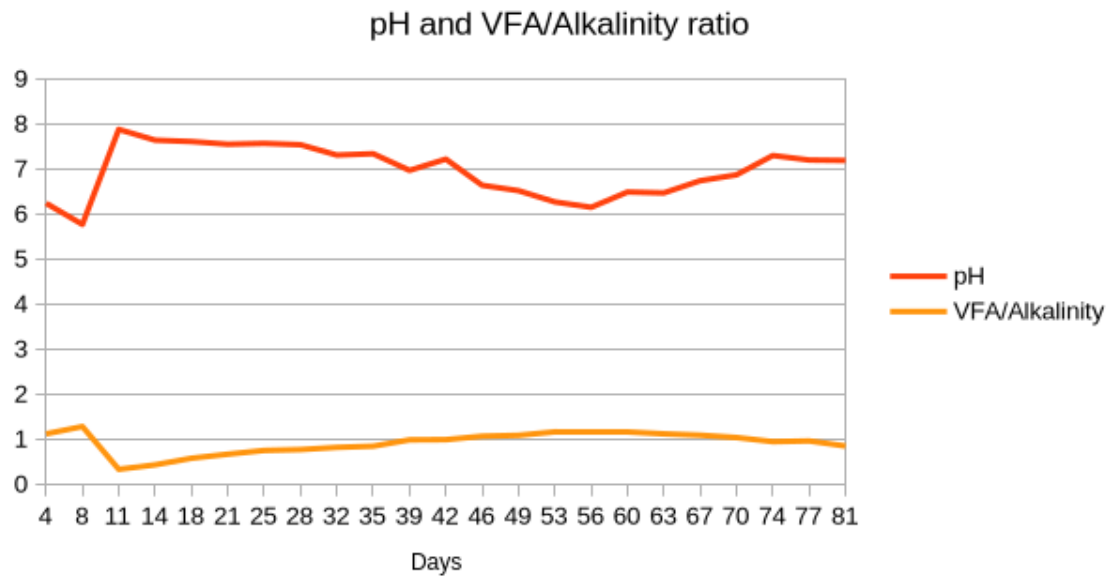
5.2.6 Alkalinity and Volatile Fatty Acids

The total alkalinity and VFA concentrations were monitored throughout the experiment to assess process stability. The results are shown in graph 5.1. Initially, both alkalinity and VFA were low, reflecting the early adaptation phase before buffer addition. After the manual introduction of KHCO_3 , alkalinity increased sharply to values above 7000 mg CaCO_3/L and remained relatively high until close to the end. This high alkalinity was maintained by the addition of bicarbonate to the feed, which helped to counteract the acidity produced during the degradation of the feed.

VFA concentrations also increased after the start of feeding, reaching values between 3000 and 5000 mg $\text{CH}_3\text{COOH}/\text{L}$. Such high VFA levels are often associated with process imbalance, as they can inhibit methanogenesis. However, the high alkalinity provided a strong buffering capacity, and the pH remained above 6.5 for most of the experiment, with the co-digestion the pH was falling and with glycerol only the pH was increasing. The VFA/alkalinity ratio is a more reliable indicator of instability. In this experiment, the ratio fluctuated between 0.3 and 1.2. While using the co-digestion feed a ration around 0.8 to 1.0 had the highest production.



Graph 5.1: Source: Author

**Graph 5.2:** Source: Author

5.2.7 Substrate used as feed

Table 5.1 shows all substrates used to feed the reactor and when they were made.

Table 5.1: Substrates used during the process

Day	Substrate 1	Substrate 2	Substrate 3	Extra Addition	COV & TS%	Volume Made
0 ^a	Food Waste, 156 g	Glucose, 1.9 g		Calcium Bicarbonate, 5 g	1.5 & 15	500 mL
4	Crude Glicerol, ≈23.2 mL	Pure Glicerol ^b , ≈31.5 mL		Potassium Bicarbonate, ≈1.9 g	0.8 & 10	250 mL
11	Food Waste, ≈12.54 g	Crude Glicerol, ≈9.02 mL	Pure Glicerol, ≈2.6 mL	Potassium Bicarbonate, ≈1.57 g	0.13-0.4 & 5-10	250 mL
18	Food Waste, ≈10.88 g	Crude Glicerol, ≈2.89 mL	Pure Glicerol, ≈1.44 mL		0.0256 & ≈2	250 mL
25	Food Waste, ≈31.12 g	Crude Glicerol, ≈44 mL	Pure Glicerol, ≈29 mL		≈1.4 & ≈10	250 mL

Table 5.1 continued

Day	Substrate 1	Substrate 2	Substrate 3	Extra Addition	COV & TS%	Volume Made
32	Food Waste, ≈ 31.01 g	Crude Glicerol, ≈ 44 mL	Pure Glicerol, ≈ 28.9 mL		≈ 1.4 & ≈ 10	250 mL
39	Food Waste, ≈ 46.52 g	Crude Glicerol, ≈ 66 mL	Pure Glicerol, ≈ 43 mL	Potassium Bicarbonate, ≈ 2.6 g	≈ 3.1 & ≈ 15	250 mL
46	Food Waste, ≈ 50.07 g	Crude Glicerol, ≈ 66 mL	Pure Glicerol, 43 mL	Potassium Bicarbonate, ≈ 2.67 g	≈ 3.16 & ≈ 15	250 mL
53 ^c	Crude Glicerol, ≈ 55 mL	Pure Glicerol, ≈ 11.5 mL		Potassium Bicarbonate, ≈ 5.38 g	0.42 & ≈ 10	500 mL
60	Crude Glicerol, ≈ 55 mL	Pure Glicerol, ≈ 12 mL		Potassium Bicarbonate, ≈ 5.18 g	0.42 & ≈ 10	500 mL ^d
67	Crude Glicerol, ≈ 41 mL	Pure Glicerol, ≈ 9 mL		Potassium Bicarbonate, ≈ 2.56 g	0.95 & ≈ 15	250 mL
74	Crude Glicerol, ≈ 60 mL	Pure Glicerol, ≈ 12.5 mL		Potassium Bicarbonate, ≈ 2.57 g	≈ 1.37 & ≈ 15	250 mL

^a It only started feeding on day 2.

^b The solution of pure glicerol was diluted 100X.

^c Change of substrate.

^d There was a leak on the feeding tube.

Source: Author

Chapter 6

Conclusion

A direct comparison of the co-digestion and glycerol only phases reveals that the co-digestion of food waste and glycerol resulted in higher biogas yields and more stable operation. The pH was decreasing faster with the co-digestion, where with glycerol only phase it wasn't decreasing that fast, and was even growing with the potassium bicarbonate that was added to maintain the pH. These results support the hypothesis that glycerol, when co-digested with food waste, can enhance biogas production by balancing the C/N ratio and providing readily available carbon, whereas glycerol alone may work, but would need a different setup and adjustments.

Using crude glycerol, with some pure glycerol to make the necessary adjustment to balance the C/N ratio, could be a viable solution for the glycerol problem. It's only major problem, is that crude glycerol contains a lot of impurities that can affect the anaerobic digestion, and for a system that can reliably consume it, it will need a better control of it's substrate.

Future studies could involve on how to use better the crude glycerol as a feed, to try subside it's downside of it's impurities and need for an extra carbon rich substrate to balance the feed.

References

- Albizzati, P. F., Tonini, D., & Astrup, T. F. (2021). A quantitative sustainability assessment of food waste management in the European Union. *Environmental Science & Technology*, 55(23), 16099–16110. <https://doi.org/10.1021/acs.est.1c03940>
- Alqudah, S. A. M. (2025). *Optimizing biogas production via ozone sludge pretreatment*.
- Anand, P., & Saxena, R. K. (2012). A comparative study of solvent-assisted pretreatment of biodiesel derived crude glycerol on growth and 1,3-propanediol production from *Citrobacter freundii* [Industrial Biotechnology]. *New Biotechnology*, 29(2), 199–205. <https://doi.org/10.1016/j.nbt.2011.05.010>
- Attarbach, T., Kingsley, M. D., & Spallina, V. (2023). New trends on crude glycerol purification: A review. *Fuel*, 340, 127485. <https://doi.org/10.1016/j.fuel.2023.127485>
- Baird, R. B., Eaton, A. D., Rice, E. W., & Bridgewater, L. L. (Eds.). (2017). *Standard methods for the examination of water and wastewater* (23rd). American Public Health Association, American Water Works Association, Water Environment Federation. <https://doi.org/10.2105/SMWW.2882.002>
- Choorit, W., & Wisarnwan, P. (2007). Effect of temperature on the anaerobic digestion of palm oil mill effluent. *Electronic Journal of Biotechnology*, 10, 376–385. <https://doi.org/10.2225/vol10-issue3-fulltext-7>
- Degueurce, A., Picard, S., Peu, P., & Trémier, A. (2020). Storage of food waste: Variations of physical–chemical characteristics and consequences on biomethane potential. *Waste and Biomass Valorization*, 11(6), 2441–2454. <https://doi.org/10.1007/s12649-018-00570-0>
- Garlapati, V. K., Shankar, U., & Budhiraja, A. (2016). Bioconversion technologies of crude glycerol to value added industrial products. *Biotechnology Reports*, 9, 9–14. <https://doi.org/10.1016/j.btre.2015.11.002>
- Gebreegziabher, B. W., Dubale, A. A., Adaramola, M. S., & Morken, J. (2025). Advancing anaerobic digestion of biodiesel byproducts: A comprehensive review. *BioEnergy Research*, 18(1), 15. <https://doi.org/10.1007/s12155-025-10820-4>

- Li, Y., Park, S. Y., & Zhu, J. (2011). Solid-state anaerobic digestion for methane production from organic waste. *Renewable and Sustainable Energy Reviews*, 15(1), 821–826. <https://doi.org/10.1016/j.rser.2010.07.042>
- Posmanik, R., Kim, A. H., Labatut, R. A., Usack, J. G., & Angenent, L. T. (2020). Granular sludge is a preferable inoculum for the biochemical methane potential assay for two complex substrates. *Bioresource Technology*, 309, 123359. <https://doi.org/10.1016/j.biortech.2020.123359>
- Ribas, M. M. F., Moraes, E. d. M., & Foresti, E. (2007). Avaliação da acurácia de diversos métodos para determinação de ácidos graxos voláteis e alcalinidade a bicarbonato para monitoramento de reatores anaeróbios. *Engenharia Sanitaria e Ambiental*, 12(3), 240–246. <https://doi.org/10.1590/S1413-41522007000300002>
- Scherhauer, S., Moates, G., Hartikainen, H., Waldron, K., & Obersteiner, G. (2018). Environmental impacts of food waste in Europe. *Waste Management*, 77, 98–113. <https://doi.org/10.1016/j.wasman.2018.04.038>
- Tomczak, W., Żak, S., Kujawska, A., & Szwast, M. (2025). The use of crude glycerol as a co-substrate for anaerobic digestion. *Molecules*, 30(17), 3655. <https://doi.org/10.3390/molecules30173655>